
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

SPECTRAL REFLECTANCE

SPECTRAL REFLECTANCE

- **The average amount of incident radiation reflected by a surface at some wavelength interval is called '*Spectral reflectance* ρ_λ" and normally expressed as a percentage**

$$\rho_\lambda = E_R(\lambda) / E_I(\lambda)$$

$$= \frac{\text{Energy of wavelength } \lambda \text{ reflected from the object}}{\text{Energy of wavelength } \lambda \text{ incident upon the object}} \times 100$$

SPECTRAL REFLECTANCE CURVE /PROFILE

- **Graph of spectral reflectance of an object as a function of wavelength is termed as "Spectral reflectance curve/profile"**
- **Curve gives us insight into the spectral characteristics of an object and has a strong influence on the choice of wavelengths in which remote sensing data is acquired for particular applications**
- **Reflectance curves are made for the optical part of the electromagnetic spectrum (up to 2.5-3 μm)**

SPECTRAL REFLECTANCE



- **To relate different kinds of surfaces and their spectral signatures (to how they look like on a satellite image), efforts are made to collect and store of typical curves in spectral libraries**
- **Reflectance measurements can be carried out in a laboratory or in the field using a Field Spectrometer (Spectroradiometer)**

SPECTRAL RESPONSE

- A basic assumption of RS is that specific surfaces have individual manner of interacting with the incident radiations, and how the surface behave is described as " SPECTRAL RESPONSE" of that surface
- Surfaces such as Soils of different types, Water with varying degrees of impurities and depths, Rocks of different lithology and Vegetation of various species have different spectral response
- Spectral response may very from time to time during year such as vegetation (crops) grows

SPECTRAL REFLECTANCE OF WATER

INTERACTION OF EMR WITH WATER

- **Possible reflectance from Water bodies**
  - **Surface reflectance**
  - **Volume reflectance**
  - **Bottom reflectance**

INTERACTION OF EMR WITH WATER

- **Spectral reflectance of water is effected by**
  - **The wavelength**
  - **Depth of the water body**
  - **Presence and concentration of dissolved matter**
  - **Surface conditions (calm/smooth, rough)**
  - **Angle of observation and illumination**

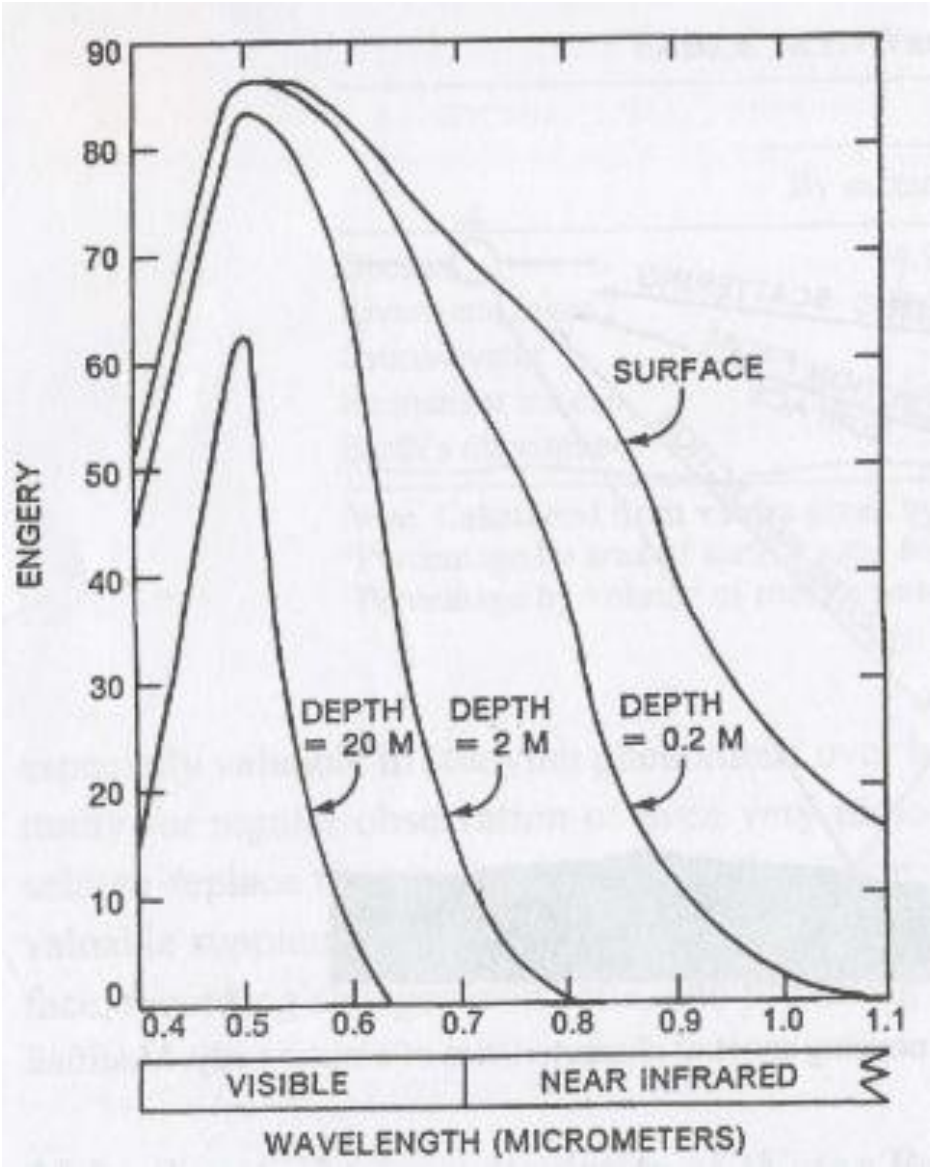
INTERACTION OF EMR WITH WATER

- Surface reflectance carry less info about water and more about the surface conditions, such as wind and waves effects
- Spectral reflectance of water body depends upon energy scattered or reflected with in the water (occurs at a range of depths)
- Deep water bodies with absence of impurities will appear blue/green (scattering by particles that are small relative to the wavelength, cause shorter WL to scatter more)

INTERACTION OF EMR WITH WATER

- In clear water, it is possible to estimate the depth from the intensity of visible radiation particularly blue light reflected from the bottom
- At greater depth, all visible radiations are absorbed and water bodies appear dark
- In NIR region, water absorbs all the incident energy and is the only natural material with this property, so water bodies can be distinguished easily from other surfaces in this part of EM spectrum

INTERACTION OF EMR WITH WATER



Moore (1978, p. 458).

INTERACTION OF EMR WITH WATER

- Sediment's concentration change spectral properties
- Brightness increases in Visible part, no more look as a dark object
- Increase in sediment contents shift the peak reflectance from shorter WL (blue) towards green and red, change of color
- Higher concentration, color may be same as of the sediments
- Degree of scattering is proportional to the concentration of particles

Sediment's Concentration

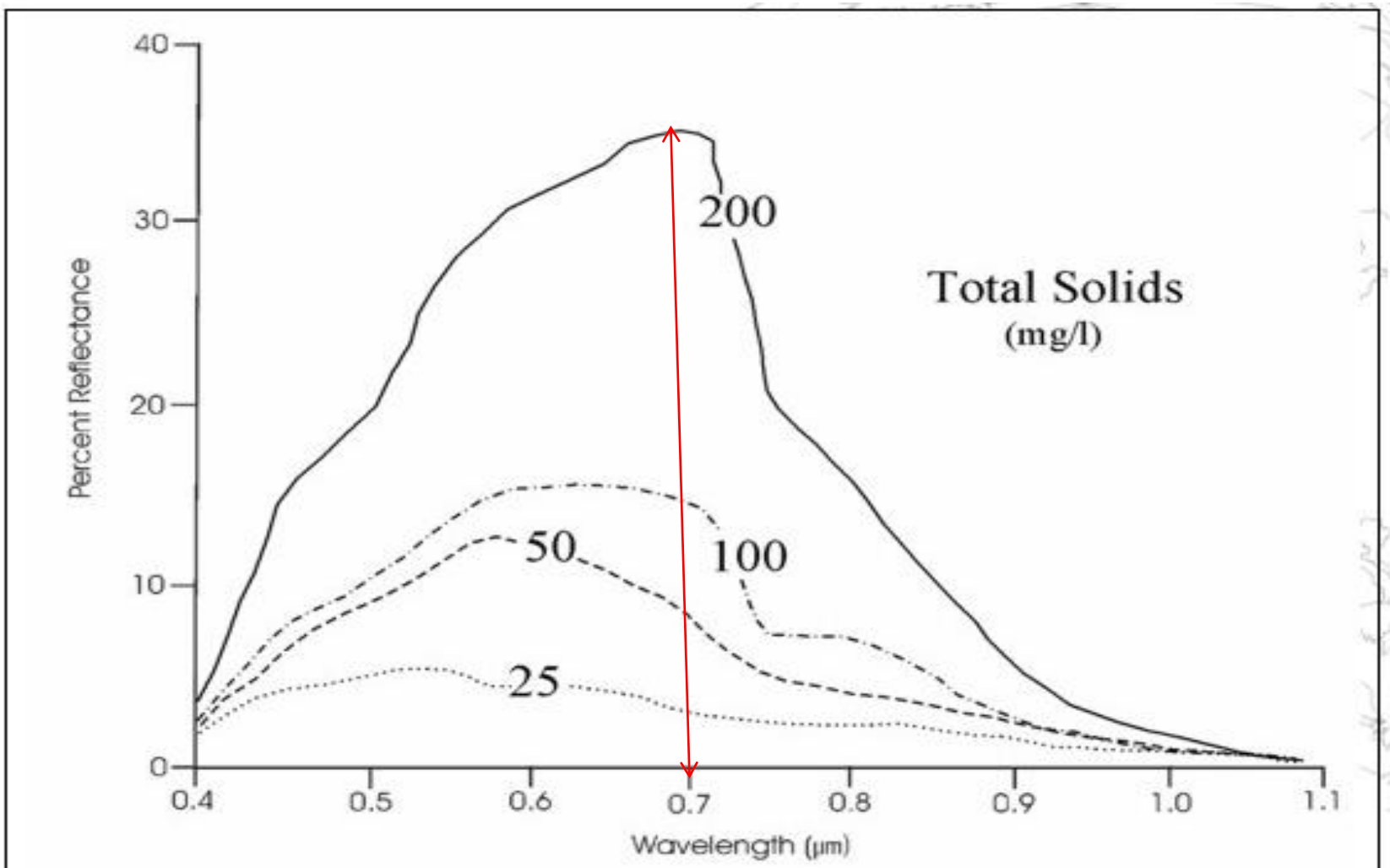


Fig 9: Variation in the spectral reflectance characteristics of turbid water according to the content of suspended solids.

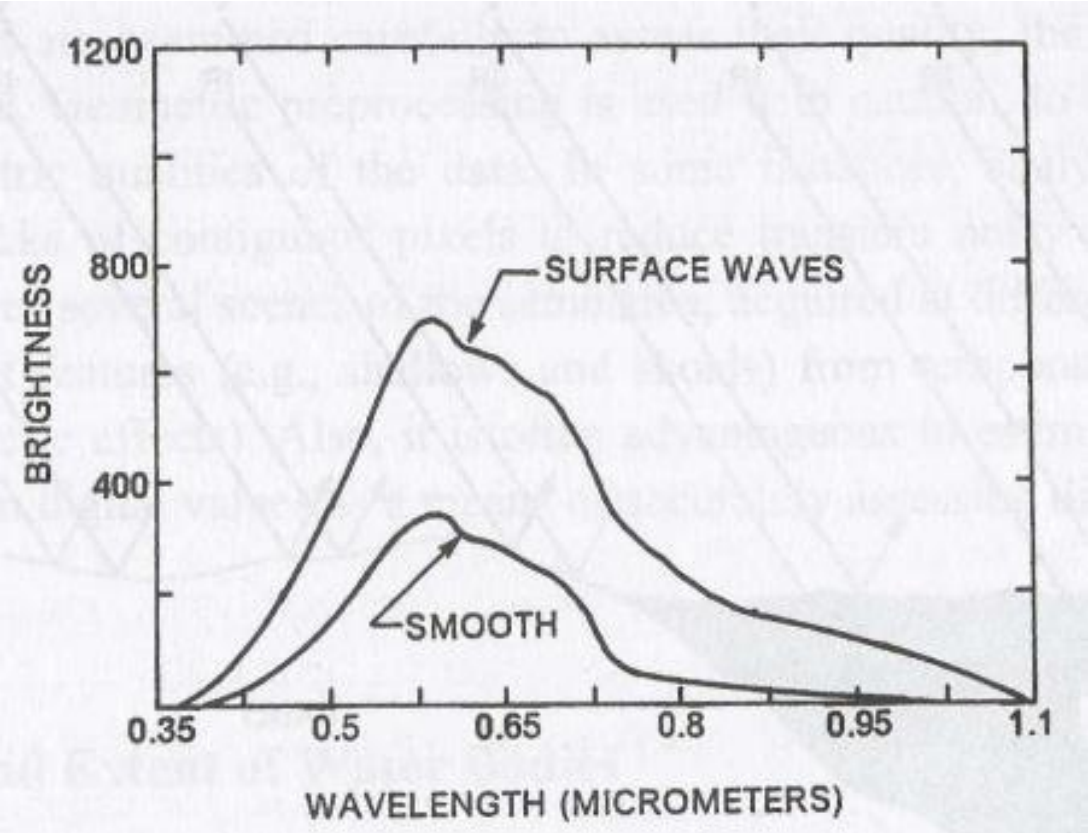
Flood Water



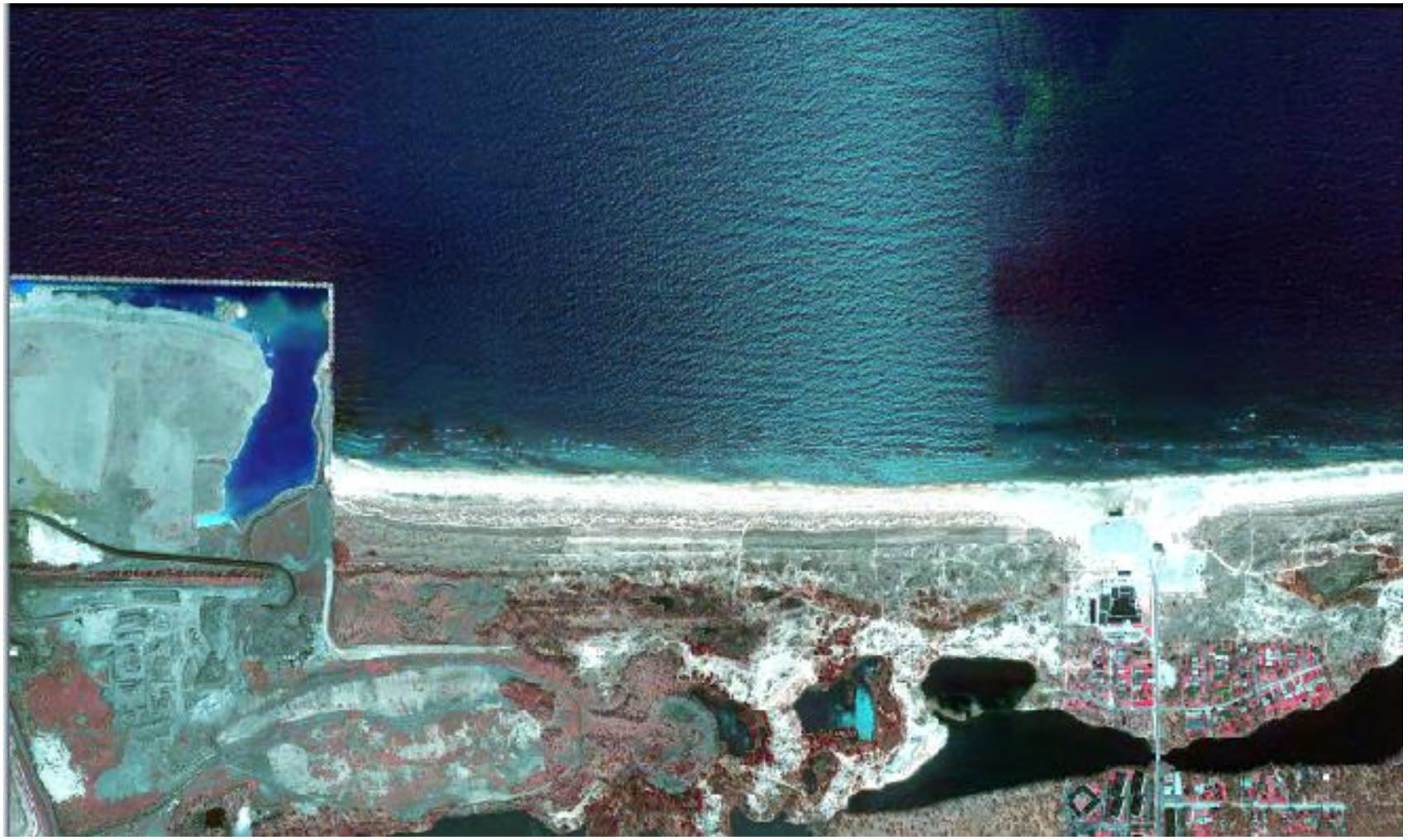


Roughness of Water Surface

- Roughness due to wave action increases the brightness (more in visible)
- Calm, smooth water surfaces direct only volume-reflected radiation to the sensor



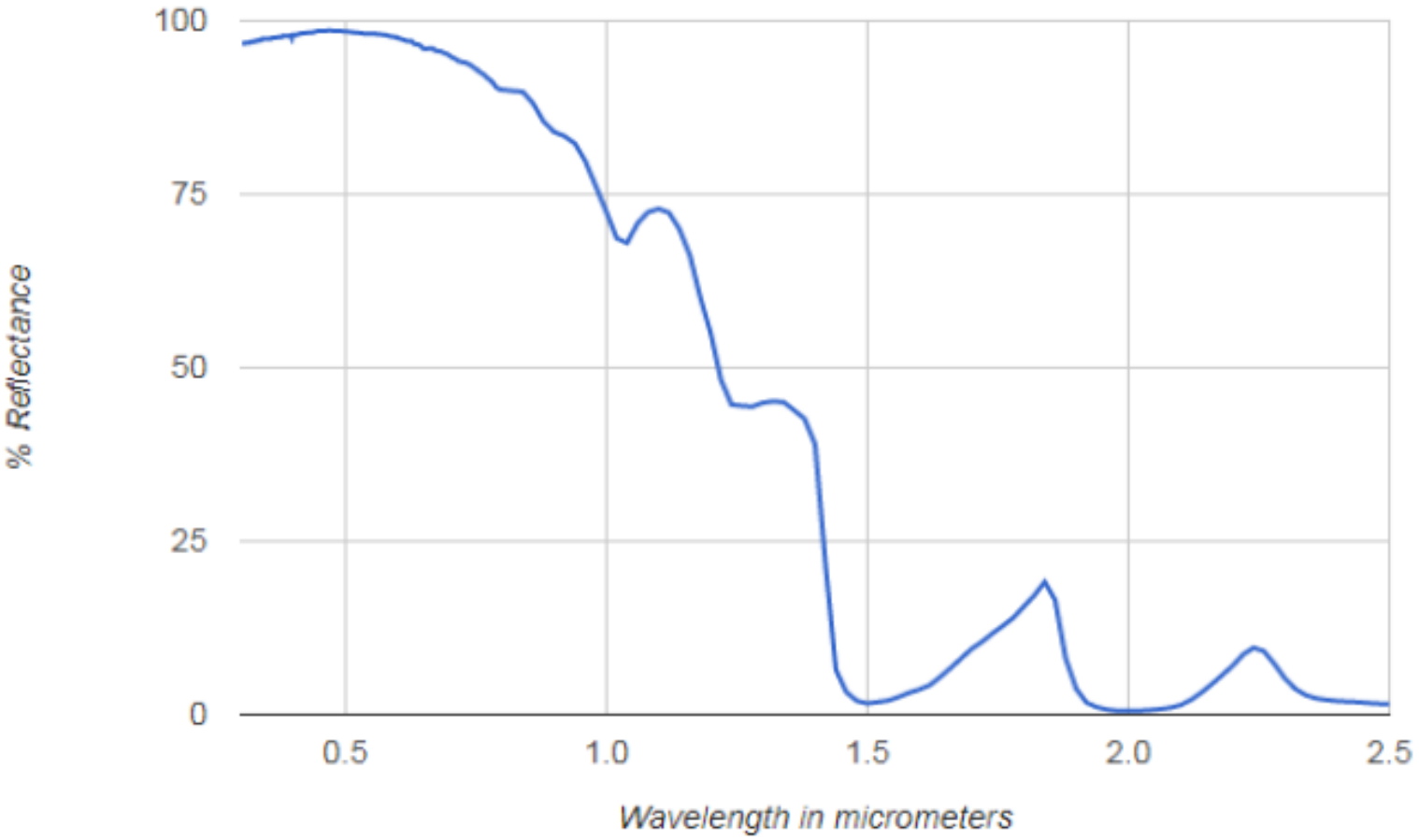
Wind and Wave Effects on Water Surface Reflectance



Wind and Wave Effects on Water Surface Reflectance



Spectral Reflectance of Water



CLEAR WATER

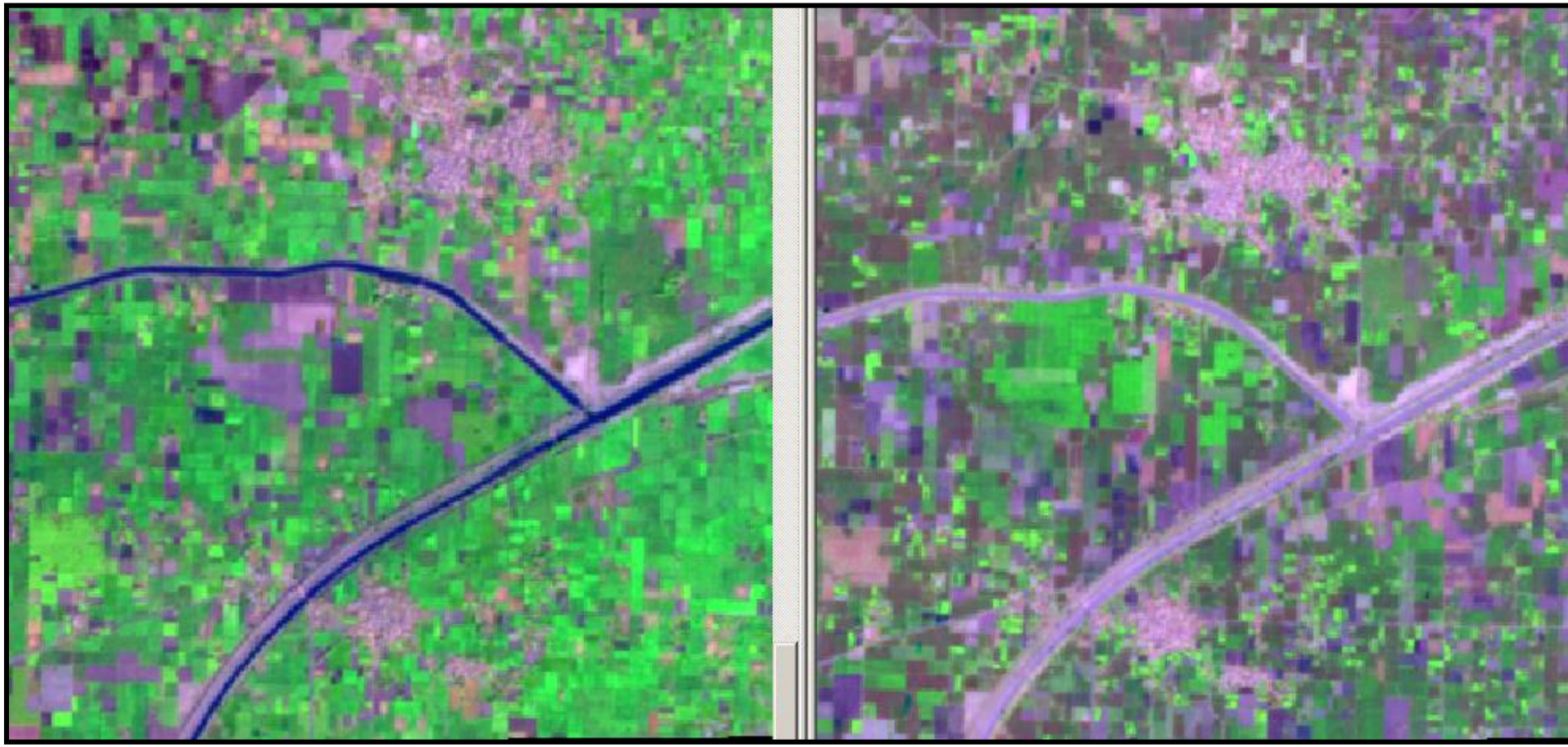


Spectral Reflectance –Temporal Effects

Apr 2005



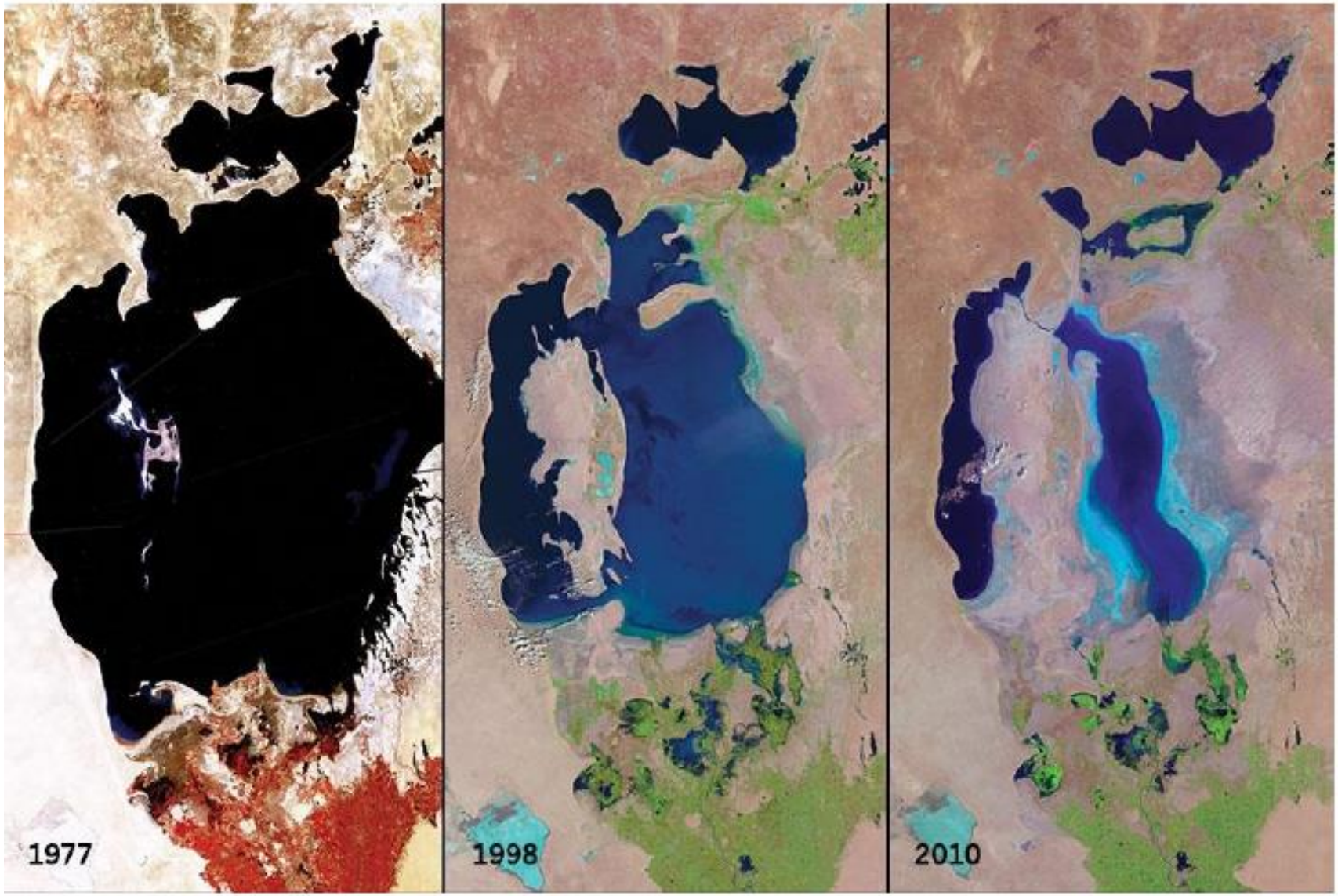
Spectral Reflectance –Temporal Effects



September **April**

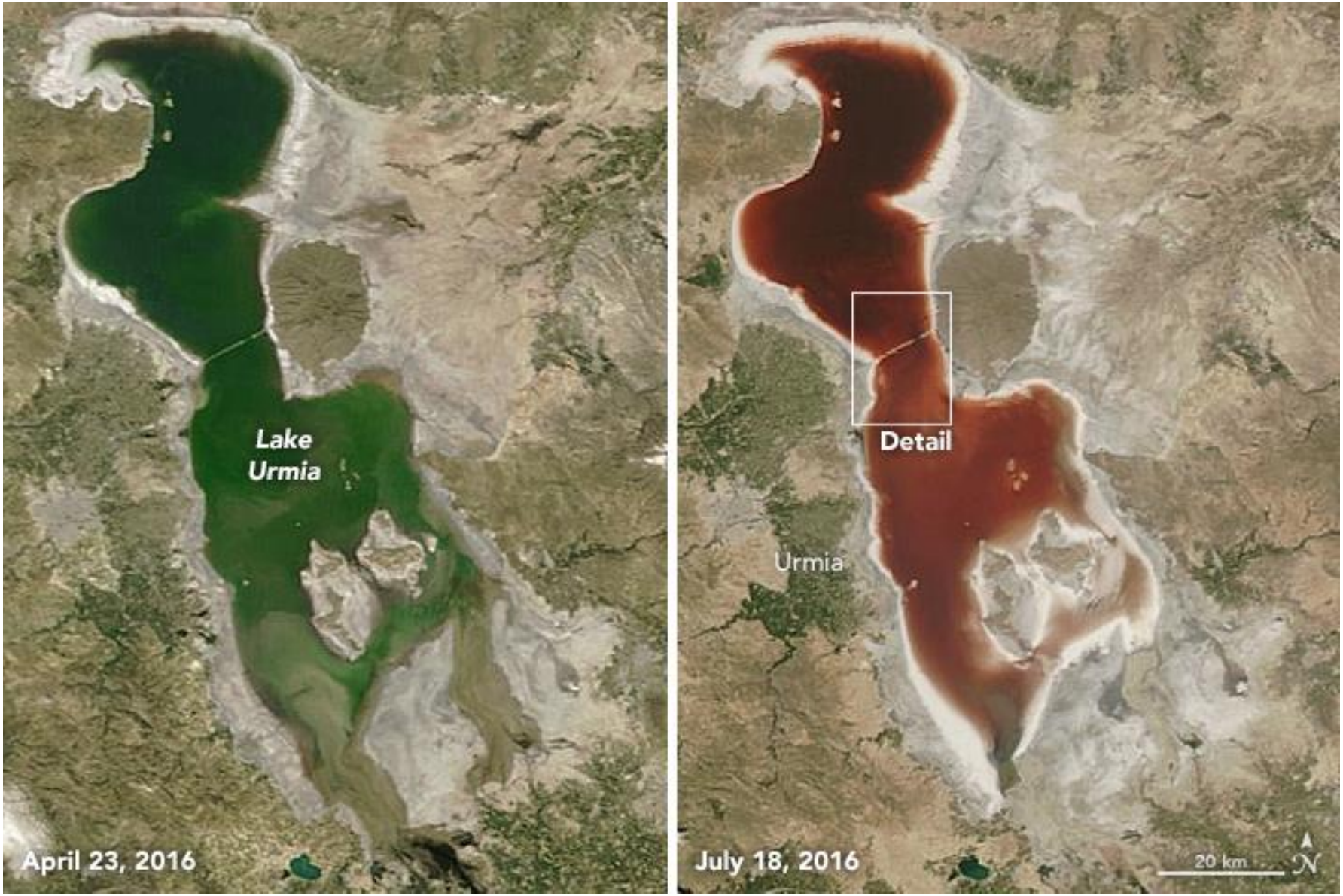
Dipalpur Canal provides water during summer only

Spectral Reflectance –Temporal Effects

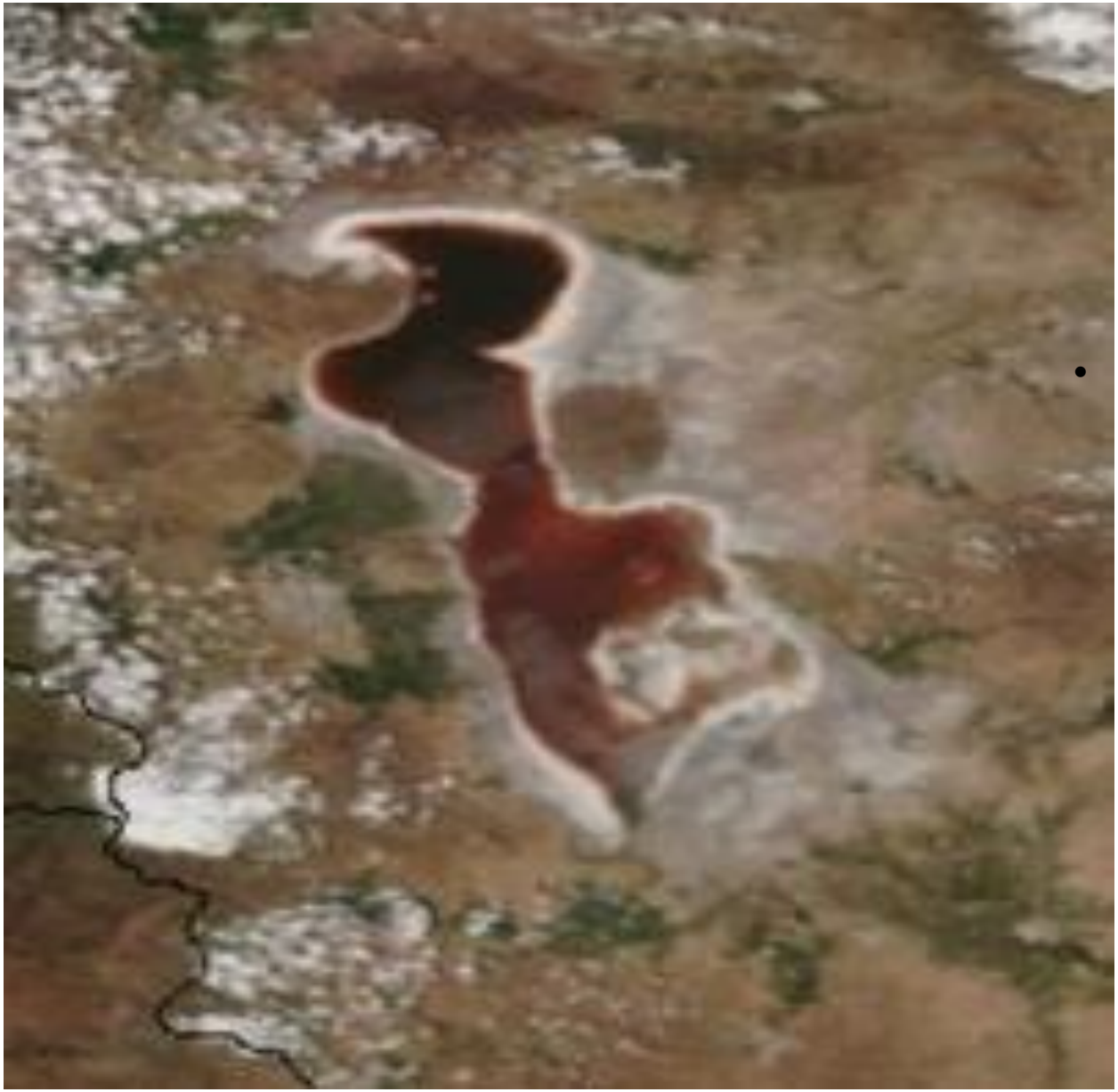




The lake, fed by rivers and streams filled with high levels of minerals and salts. Because the lake lacks an outflow, water can only leave by evaporation. The salts in the water do not evaporate, but accumulate as crusty, reflective mineral deposits on the lake shore. These can be seen from space as a white rim around the lake.


Lake
Urmia
April 23, 2016
Detail
Urmia
July 18, 2016
20 km
N

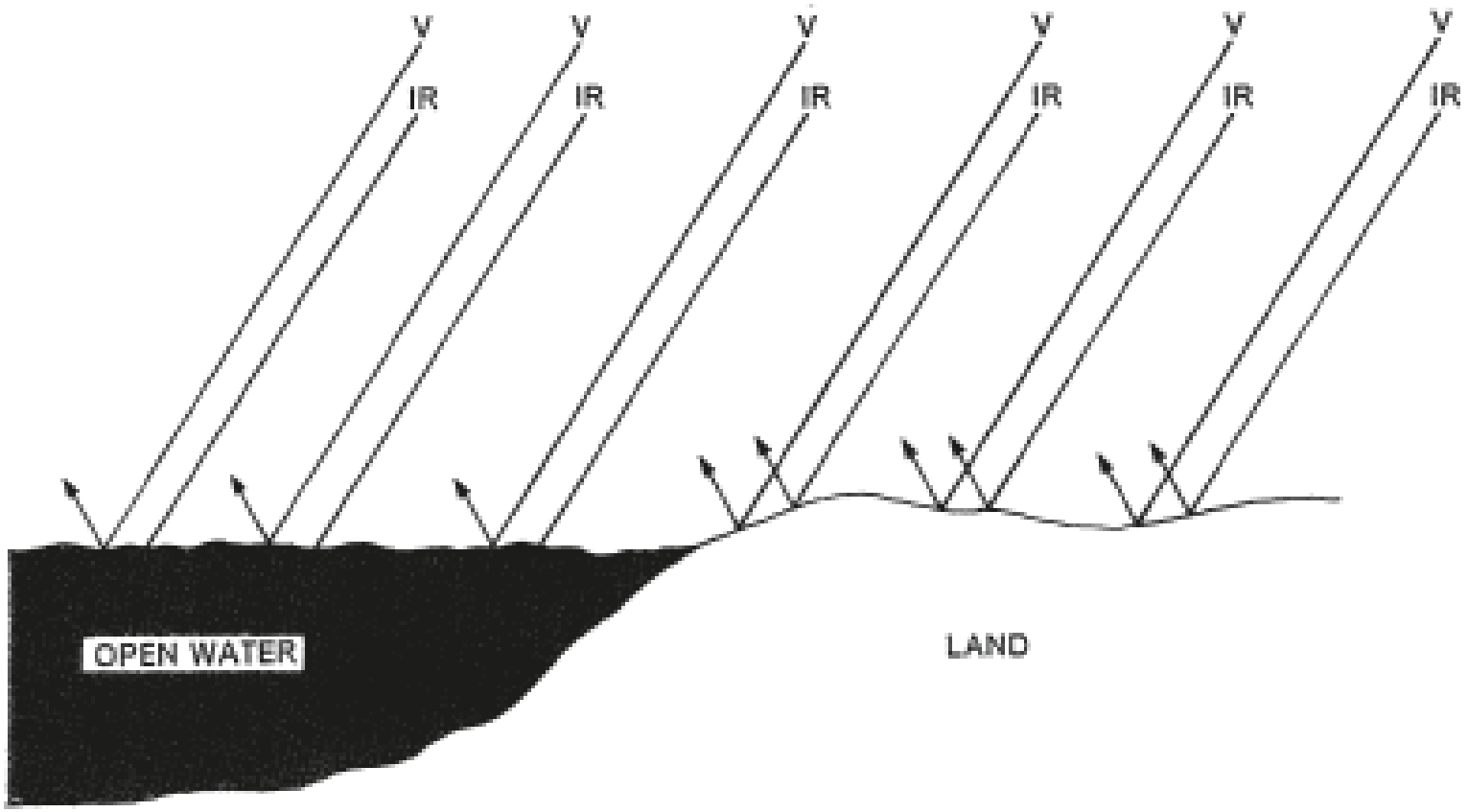
In Jul 2016, it appeared in the Satellite Imagery in Kool-Aid tropical Punch color, a drastic change on the





- Traditionally blue or greenish in color, Lake Urmia's red hue is attributed to a type of specialized bacteria, which flourish in extreme salty conditions Called halophiles (“salt-loving”), they produce pink-red and when found in abundance, the pigments from these bacteria can turn super-salty water pink or red.

Land-Water Boundary



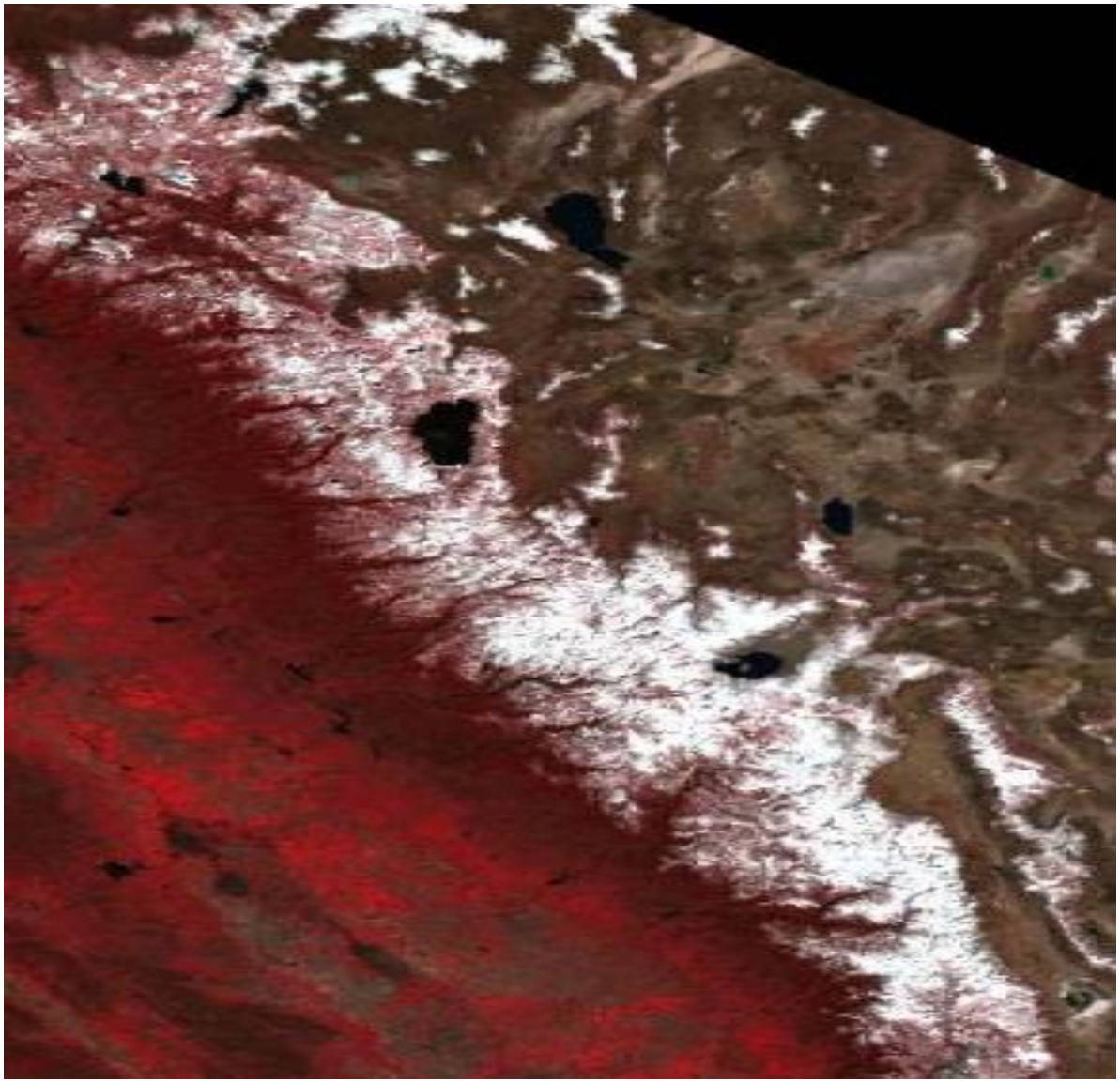
Land–Water boundary: usually easiest in the near infrared region, where land, especially if vegetated, is bright, and open water dark

SPECTRAL REFLECTANCE OF SNOW

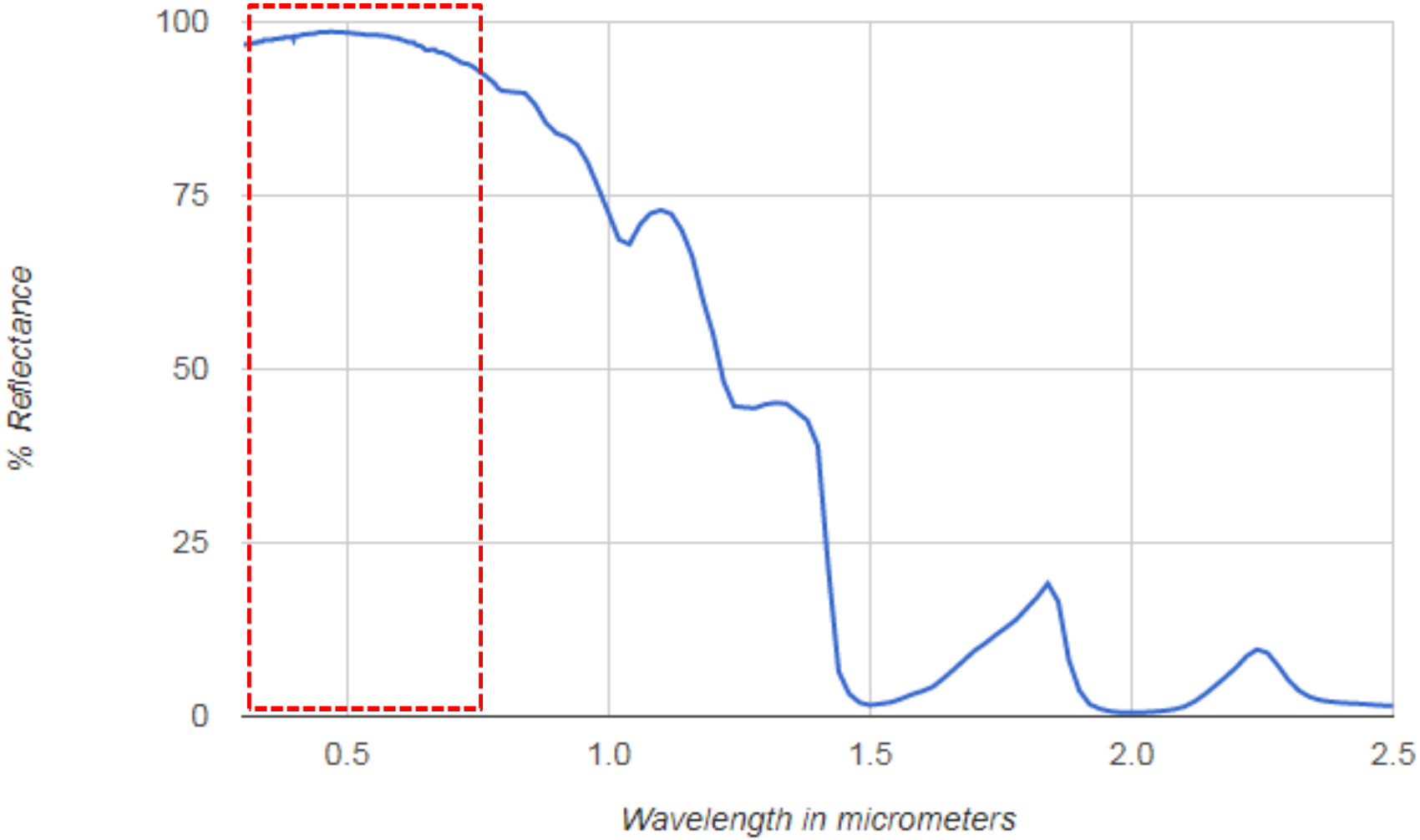
Spectral Reflectance- Snow

- **The reflectance of the snowpack depend on the**
  - **Wavelength**
  - **Age (time)**
  - **Grain size distribution of the snow**
  - **Depth and density of the snowpack**
  - **Size and amount of the impurities**

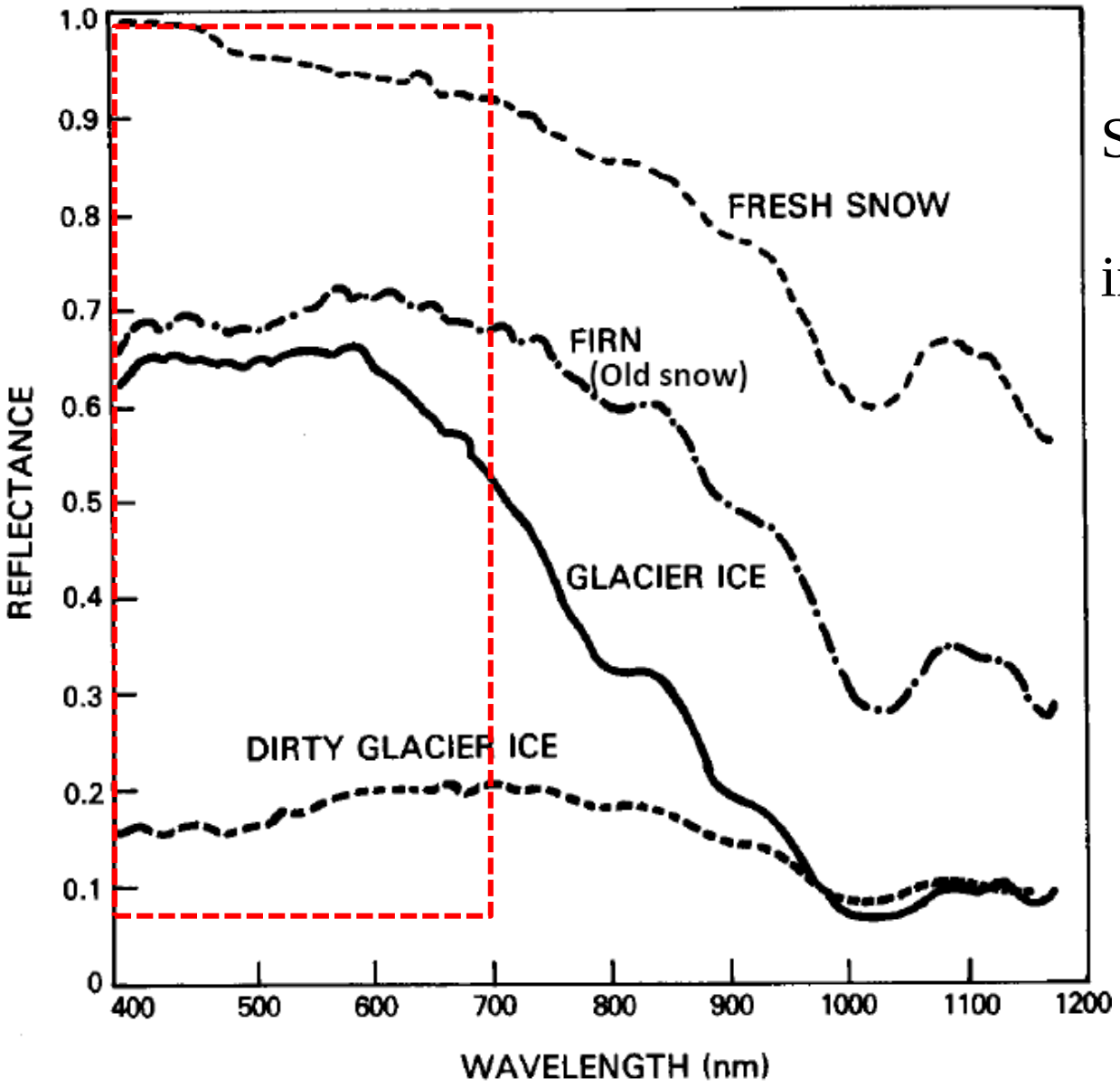
Spectral Reflectance- Snow



Spectral Reflectance of Snow



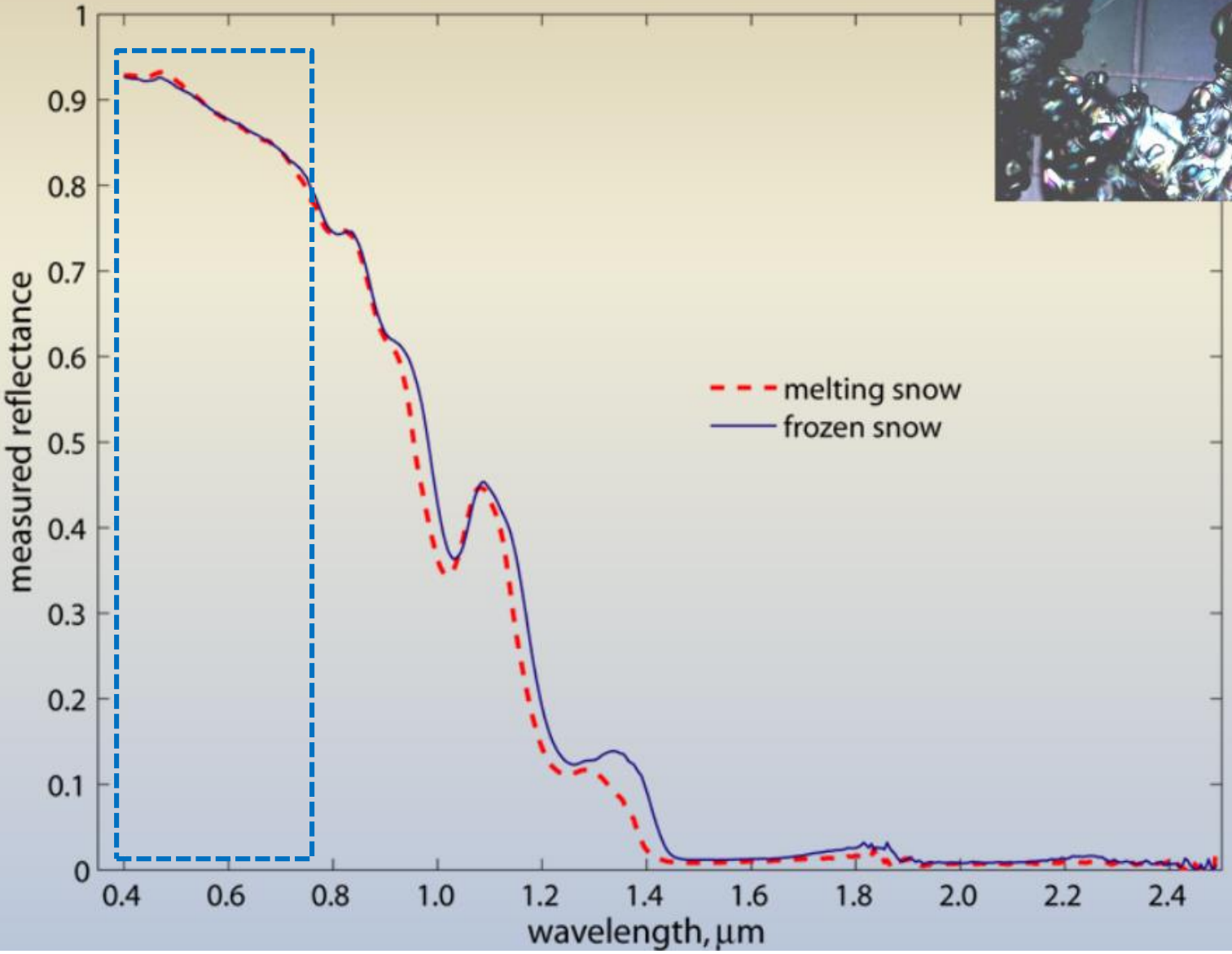
Spectral Reflectance- Age



Size and amount of the impurities

Melting Vs Frozen Snow Snow





Spectral Reflectance of Clean Snow- Grain Size

